

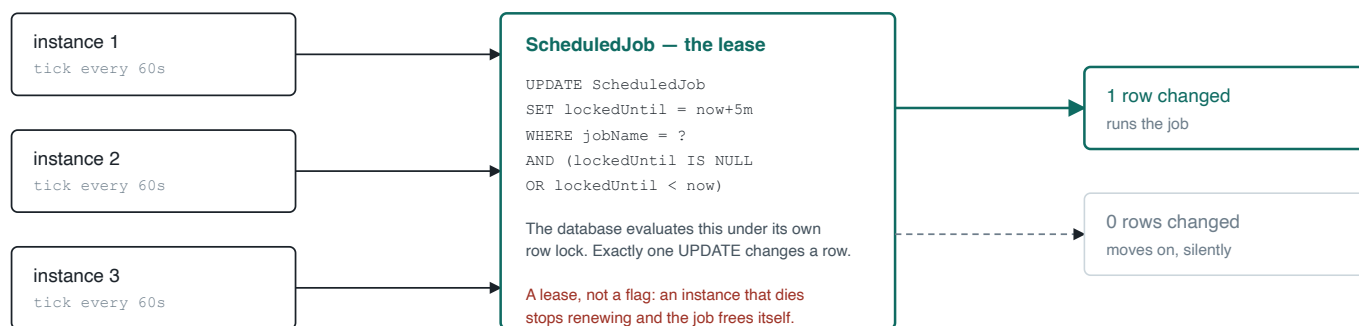
IntelliOps4 — architecture of the beta features

Five drawings covering everything added in the agentic build: background work, the proactive jobs, projection, batched approval, and questions that outlive a request. Drawn from the code as deployed on 6 September 2026.

■ background work ■ a gate a human must pass ■ a projection, not a fact ■ language model ■ SAP

1 · The scheduler, and why the lock is the feature

Cloud Foundry runs several copies of the application. A plain timer fires in every one of them, so a daily digest would be delivered three times and a threshold alert would page someone in triplicate — a fault invisible on one machine. The database decides the winner, because it is the only thing all instances share.



Every run is recorded

JobRun holds start, finish, duration, outcome and which instance ran it — for failures as much as successes.

A job that has thrown every night for a week is seven rows, not an absence of evidence.

Figure 1 — the scheduler lease. The conditional UPDATE is the whole guarantee: read-then-write would let two instances both see an expired lease and both believe they had won.

2 · What runs on its own

Four jobs, all reading through the same tools the chat uses. A digest that disagreed with what the chat says about the same data would make both untrustworthy, and the divergence would be invisible until someone checked by hand.

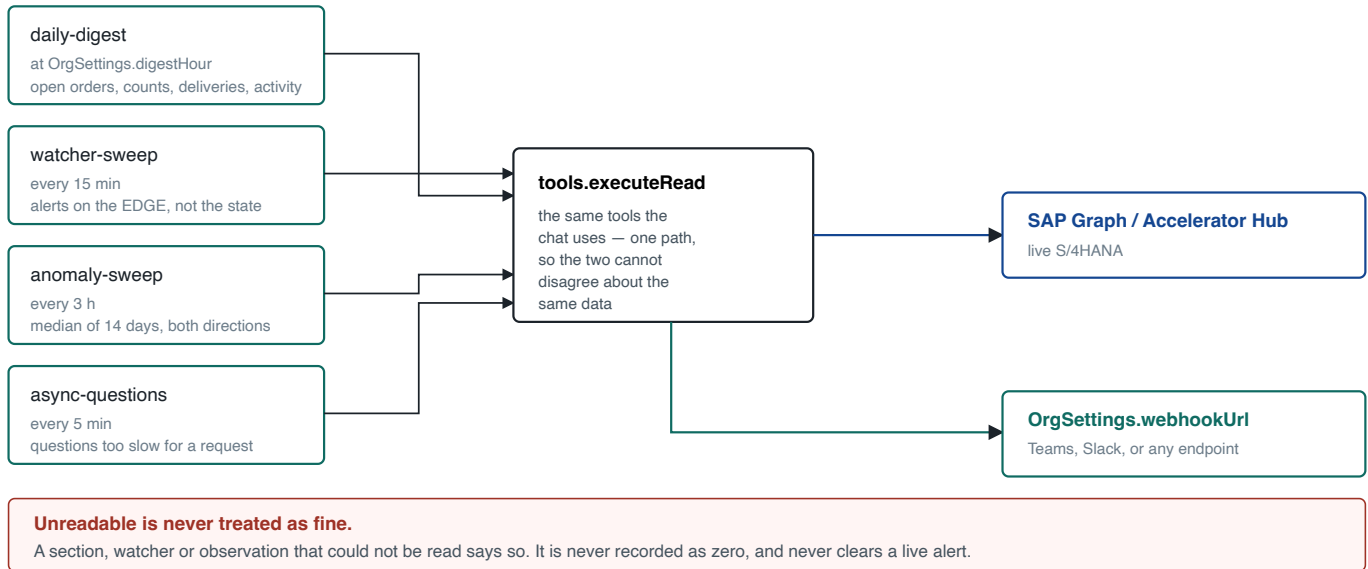


Figure 2 — the four background jobs. Every one degrades per item rather than all-or-nothing, because a failure at 6am reaches someone who is not reading logs.

3 • Three kinds of tool

The agent loop used to know two kinds of tool: a read, and a write that stops for approval. Projection is a third, and it needed to be third rather than a variant of either — its inputs are grounded and its output is not.

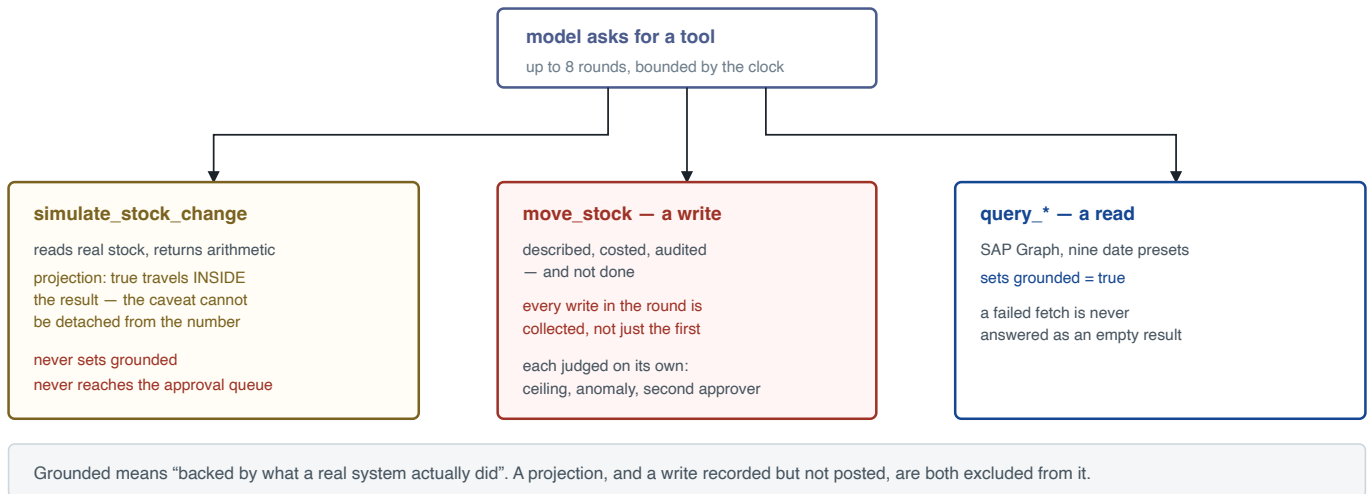


Figure 3 — three tool categories. Making projection a third branch rather than a read is what keeps a computed figure from inheriting the badge that certifies real ones.

4 • Batched approval

One question can propose several writes. The loop used to return on the first and discard the rest, so a user approved one card and had every reason to believe all five had happened.

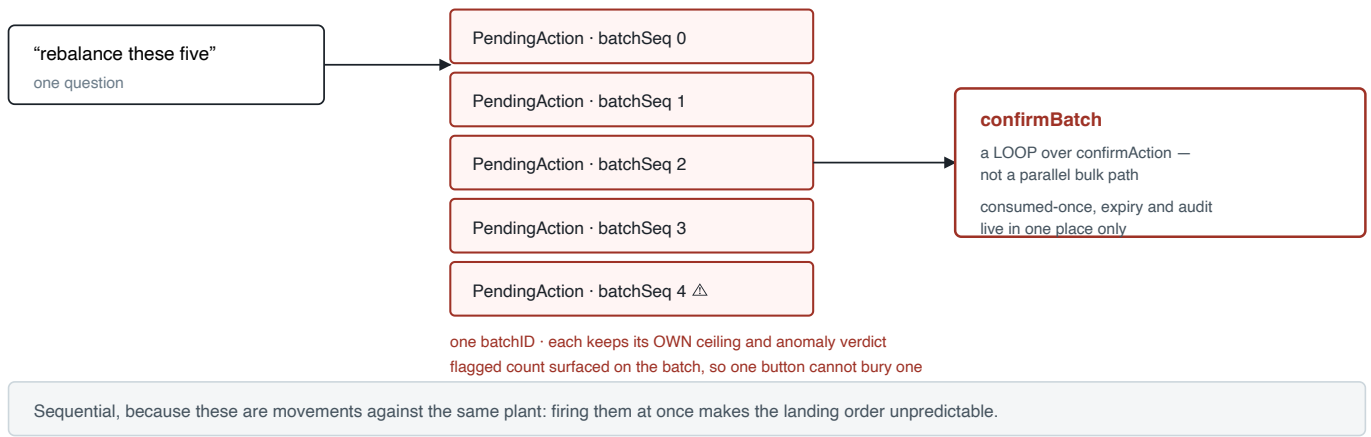


Figure 4 — batched approval. A second bulk implementation would be a second place for consumed-once, expiry and audit to be true, and therefore a second place for them to stop being true.

5 · A question that outlives its request

Every synchronous question is bounded at about 75 seconds because a browser is waiting and the gateway gives up at two minutes. A root-cause chain across a month of movements is legitimately slower than that.

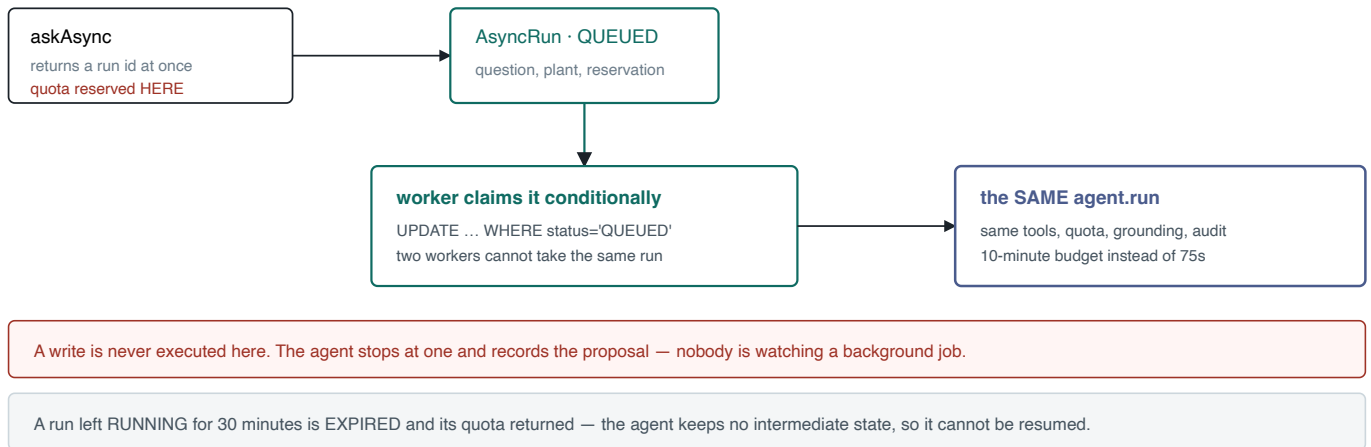


Figure 5 — asynchronous questions. The worker calls the same agent as the request path; a second implementation would be a second place for grounding, quota and audit to stop being true.